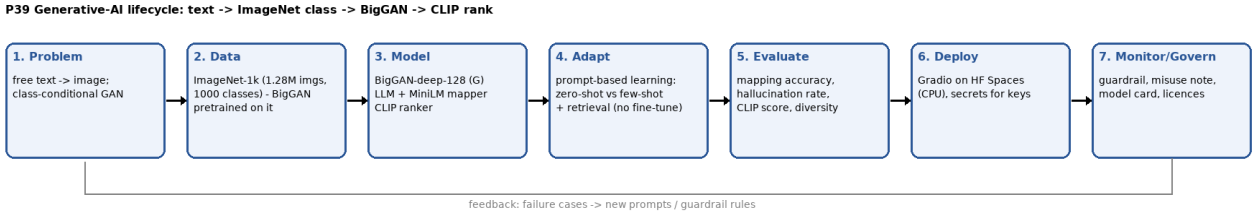
P39 · Class-Conditional Image Studio (BigGAN)

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Course: Generative AI, CA mini-project

Live app: <live app URL>
GitHub: <GitHub repo URL>

Goal: generate images from free text via the nearest ImageNet class. Pipeline: guardrail → LLM (gpt-oss-20b) guesses class names → MiniLM retrieves candidate classes → LLM (few-shot) picks one → BigGAN-deep-128 generates with truncation ψ → CLIP ranks samples against the text.

Headline results: text→class accuracy 47% (embedding) → 87% (few-shot v1) → **93–97%** (few-shot + query expansion, two runs), 0% invalid answers vs 27% for zero-shot; truncation 0.2→1.0 raises sample diversity ~4×; CLIP score BigGAN 28.5 vs sd-turbo 32.2, while BigGAN is ~18× faster on CPU.



U1 · Foundations: truncation as sampling from a narrower distribution

BigGAN's generator G(z, y) maps a latent $z \in R^{128}$ and a one-hot class y to an image. During training z ~ N(0, I). The **truncation trick** samples z from a normal truncated to [−2, 2] and scales it by $\psi \leq 1$, so z lies in a narrower region around the mode of the prior. Images near the mode are the most "typical" for the class, which gives higher fidelity, while the lower spread means less diversity. This is a precision/recall trade-off controlled at sampling time, with no retraining.

Generative vs discriminative: CLIP and the sentence encoder are used discriminatively (scoring p(match | image, text)); BigGAN is generative (sampling x ~ p(x | y)). **Prompt techniques compared:** zero-shot ("name the ImageNet class") vs few-shot with a retrieved candidate list (see U2 table).

Truncation trade-off: mean CLIP score and CLIP-embedding diversity (8 samples per cell)

Prompt	Class	CLIP @ 0.2	CLIP @ 0.5	CLIP @ 1.0	Div @ 0.2	Div @ 0.5	Div @ 1.0
a friendly family dog with golden fur	golden retriever	29.6	29.4	29.1	0.025	0.049	0.084
a mountain erupting with lava	volcano	26.8	26.0	26.0	0.021	0.063	0.145
a slice of italian food with cheese and tomato	pizza	27.7	27.7	27.7	0.029	0.043	0.093
an old wooden ship with pirate flags	schooner	26.5	26.8	26.4	0.028	0.045	0.108
an orange and black butterfly that migrates	monarch	31.2	30.6	30.6	0.013	0.026	0.076

Prompt	Class	CLIP @ 0.2	CLIP @ 0.5	CLIP @ 1.0	Div @ 0.2	Div @ 0.5	Div @ 1.0
Mean		28.4	28.1	28.0	0.023	0.045	0.101

As ψ grows from 0.2 to 1.0, diversity rises roughly fourfold while the CLIP score changes little or falls slightly. This is the latent-space view of truncation.

U2 · LLM maps text → class; pretrained BigGAN generates

The LLM is used with **prompt-based learning** (no fine-tuning). Zero-shot asks for a class name, which may not exist (a hallucination). Few-shot + retrieval gives three worked examples and the top-20 classes from all-MiniLM-L6-v2. The answer must be an index from that list, so it is validated. The embedding-only nearest label is the no-LLM baseline. **Iteration:** every v1 error happened because the right class was not among the encoder's top 20 ("king of the jungle" is not near "lion" in embedding space). v2 adds **query expansion**: the LLM first guesses 3 class names and candidates are retrieved for the text and for each guess. Accuracy rose from 87% to 93–97% over two runs (28/30 and 29/30; the LLM is not fully deterministic at temperature 0), at the cost of a second LLM call. BigGAN-deep injects the class through conditional batch-norm in every residual block and uses a **self-attention** block at 64×64 so distant regions of the image stay consistent.

Mapping accuracy on 30 labelled prompts

Method	Top-1 accuracy	Hallucination rate	LLM errors	Mean latency
embedding	47% (14/30)	0%	0	16 ms
llm_zero	90% (27/30)	27%	0	874 ms
llm_fewshot_v1	87% (26/30)	3%	0	3144 ms
llm_fewshot	93% (28/30)	0%	0	6441 ms

Mapping errors

Method	Prompt	Gold	Predicted
embedding	a fluffy orange cat sleeping	tabby, tiger cat, Persian cat, Egyptian cat	Siamese cat
embedding	the king of the jungle roaring	lion	king snake
embedding	a black and white bear eating bamboo	giant panda	American black bear
embedding	a big striped cat in the forest	tiger, tiger cat	Madagascar cat
embedding	a slow reptile with a shell walking on land	box turtle, mud turtle, terrapin	alligator lizard
embedding	a bird that cannot fly and lives in Antarctica	king penguin	hummingbird
embedding	the most feared predator of the ocean	great white shark, tiger shark	sea snake
embedding	the yellow vehicle that takes kids to class	school bus	recreational vehicle
embedding	a sour green apple	Granny Smith	custard apple
embedding	a small strong coffee in a tiny cup	espresso	coffee mug
embedding	a stringed instrument played with a bow under the chin	violin	bow
embedding	an instrument with black and white keys	grand piano, upright	electric guitar
embedding	a portable computer that folds open	laptop, notebook	hand-held computer
embedding	a tower with a light that guides ships	beacon	water tower
embedding	an orange and black butterfly that migrates	monarch	sulphur butterfly
embedding	an animal that hops and carries its baby in a pouch	wallaby	beagle
llm_zero	a slow reptile with a shell walking on land	box turtle, mud turtle, terrapin	leatherback turtle
llm_zero	a sour green apple	Granny Smith	custard apple

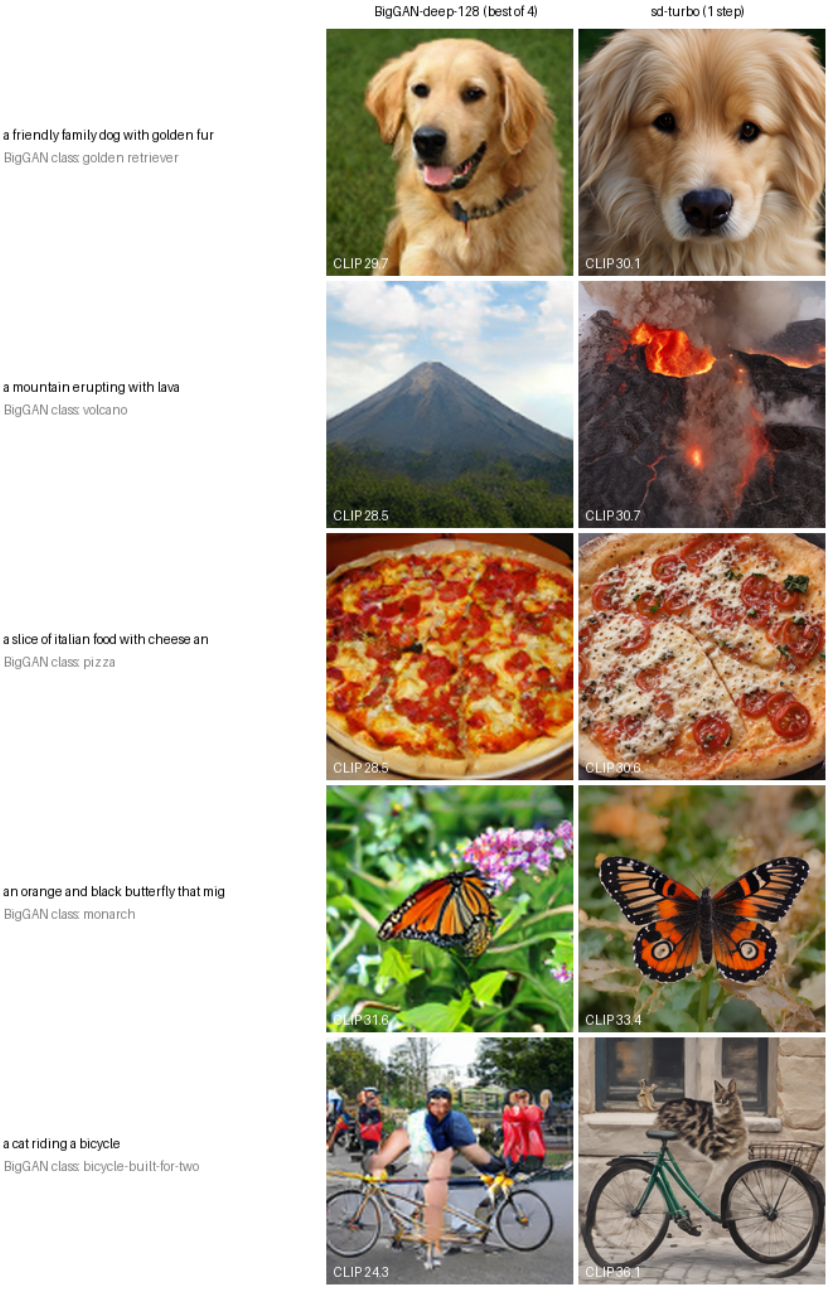
Method	Prompt	Gold	Predicted
llm_zero	an orange and black butterfly that migrates	monarch	cabbage butterfly
llm_fewshot_v1	the king of the jungle roaring	lion	tiger
llm_fewshot_v1	a sour green apple	Granny Smith	orange
llm_fewshot_v1	an orange and black butterfly that migrates	monarch	sulphur butterfly
llm_fewshot_v1	an animal that hops and carries its baby in a pouch	wallaby	beagle
llm_fewshot	a sour green apple	Granny Smith	lemon
llm_fewshot	a stringed instrument played with a bow under the chin	violin	cello

Classes chosen for the grid prompts: "a friendly family dog with golden fur" → golden retriever (llm_fewshot); "a mountain erupting with lava" → volcano (llm_fewshot); "a slice of italian food with cheese and tomato" → pizza (llm_fewshot); "an old wooden ship with pirate flags" → schooner (llm_fewshot); "an orange and black butterfly that migrates" → monarch (llm_fewshot)

U3 · CLIP ranking and comparison with diffusion

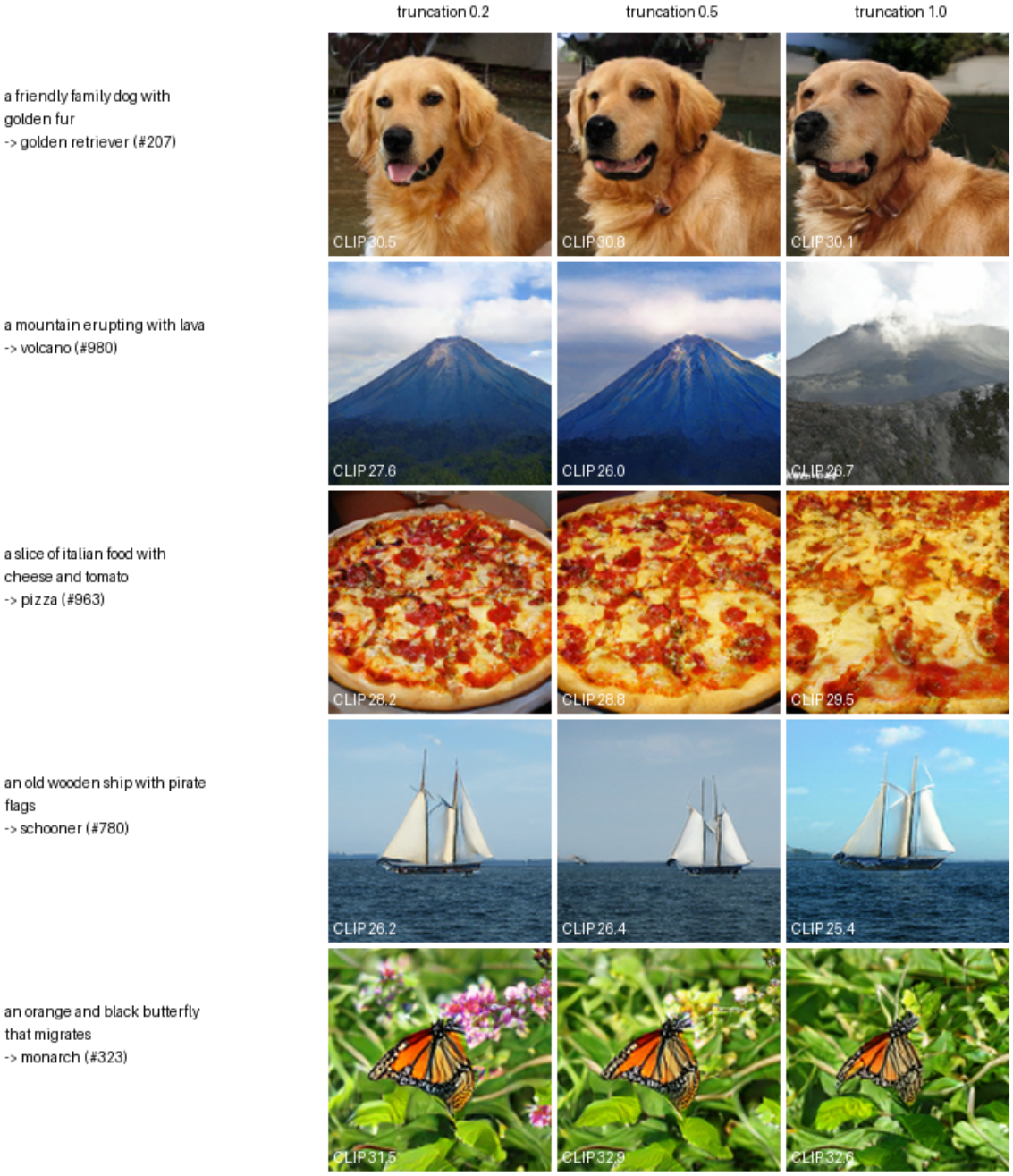
Every BigGAN sample is scored with CLIP ViT-B/32 (100 × cosine between image and prompt embeddings), and the gallery is sorted by this score. The same prompts were also rendered with sd-turbo (1-step text-to-image diffusion).

Prompt	BigGAN class	BigGAN CLIP	sd-turbo CLIP	BigGAN time	sd-turbo time
a friendly family dog with golden fur	golden retriever	29.7	30.1	8.6 s*	35.2 s
a mountain erupting with lava	volcano	28.5	30.7	1.8 s	35.5 s
a slice of italian food with cheese and tomato	pizza	28.5	30.6	1.8 s	37.9 s
an orange and black butterfly that migrates	monarch	31.6	33.4	2.0 s	36.1 s
a cat riding a bicycle	bicycle-built-for-two	24.3	36.1	1.9 s	35.1 s
Mean		28.5	32.2		



U4 · PyTorch truncation slider and grid

The Gradio app has a truncation slider (0.02–1.0), sample count and seed controls, and a *Truncation explorer* tab that renders the same seed at ψ = 0.1 … 1.0. Below is the required **5 prompts × 3 truncation values** grid (each cell = CLIP-best of 4).



U5 · Responsible AI: misuse, class imbalance, hallucination

Guardrail: input filter for violence, sexual content, hate, self-harm and real-person/deepfake requests; 200-character limit; weapon classes (assault rifle, revolver, rifle, missile …) are labelled, and blocked when combined with harmful-intent words.

Misuse note: outputs are synthetic 128 px images of generic objects. They must not be passed off as real photos or used to deceive. The app is for education and research only.

Class imbalance: ImageNet-1k has ~120 dog-breed classes but only 3 person-like classes (groom, scuba diver, ballplayer), and its photos skew towards Western web imagery. The probe below shows person prompts being forced onto unrelated objects.

Out-of-vocabulary prompts (failure cases)

Method	Prompt	Mapped to	Encoder sim	Low-confidence flag
embedding	a giraffe at sunset	polecat	0.504	no
llm_zero	a giraffe at sunset	hognose snake	0.436	no
llm_fewshot_v1	a giraffe at sunset	polecat	0.504	no
llm_fewshot	a giraffe at sunset	hognose snake	0.436	no
embedding	a dragon breathing fire	European fire salamander	0.57	no
llm_zero	a dragon breathing fire	fire engine	0.479	no
llm_fewshot_v1	a dragon breathing fire	Komodo dragon	0.482	no
llm_fewshot	a dragon breathing fire	Komodo dragon	0.482	no
embedding	freedom	padlock	0.233	yes
llm_zero	freedom	padlock	0.233	yes
llm_fewshot_v1	freedom	prison	0.201	yes
llm_fewshot	freedom	bald eagle	0.102	yes
embedding	a cat riding a bicycle	Egyptian cat	0.513	no
llm_zero	a cat riding a bicycle	bicycle-built-for-two	0.355	no
llm_fewshot_v1	a cat riding a bicycle	tabby	0.258	no
llm_fewshot	a cat riding a bicycle	bicycle-built-for-two	0.355	no
embedding	the Eiffel Tower at night	water tower	0.512	no
llm_zero	the Eiffel Tower at night	water tower	0.512	no
llm_fewshot_v1	the Eiffel Tower at night	water tower	0.512	no
llm_fewshot	the Eiffel Tower at night	water tower	0.512	no
embedding	a smartphone with a cracked screen	screen	0.551	no
llm_zero	a smartphone with a cracked screen	cellular telephone	0.457	no
llm_fewshot_v1	a smartphone with a cracked screen	cellular telephone	0.457	no
llm_fewshot	a smartphone with a cracked screen	cellular telephone	0.457	no

People / bias probe (ImageNet has almost no person classes)

Method	Prompt	Mapped to	Encoder sim	Low-confidence flag
embedding	a doctor	medicine chest	0.467	no
llm_zero	a doctor	medicine chest	0.467	no
llm_fewshot_v1	a doctor	lab coat	0.273	no
llm_fewshot	a doctor	lab coat	0.273	no
embedding	a nurse	medicine chest	0.395	no
llm_zero	a nurse	medicine chest	0.395	no
llm_fewshot_v1	a nurse	medicine chest	0.395	no
llm_fewshot	a nurse	medicine chest	0.395	no
embedding	an engineer at work	power drill	0.418	no
llm_zero	an engineer at work	carpenter's kit	0.405	no
llm_fewshot_v1	an engineer at work	power drill	0.418	no
llm_fewshot	an engineer at work	laptop	0.208	no
embedding	a CEO in a meeting	entertainment center	0.231	yes
llm_zero	a CEO in a meeting	harvestman	0.158	yes
llm_fewshot_v1	a CEO in a meeting	desk	0.185	yes
llm_fewshot	a CEO in a meeting	suit	0.051	yes
embedding	a farmer in India	tractor	0.433	no
llm_zero	a farmer in India	tractor	0.433	no
llm_fewshot_v1	a farmer in India	tractor	0.433	no
llm_fewshot	a farmer in India	tractor	0.433	no
embedding	a bride	gown	0.483	no
llm_zero	a bride	gown	0.483	no
llm_fewshot_v1	a bride	gown	0.483	no
llm_fewshot	a bride	gown	0.483	no

Failure cases: v2 still maps "a sour green apple" to *lemon* instead of *Granny Smith* (the LLM reasons on "sour"). The embedding baseline maps "an orange and black butterfly that migrates" to *sulphur butterfly* (yellow): colour words are ignored by the nearest-label search. Compositional prompts ("a cat riding a bicycle" → *bicycle-built-for-two*) cannot be expressed by a single class at all, and abstract prompts ("freedom") trigger the low-confidence warning.

Licences: BigGAN code MIT, weights from DeepMind TF-Hub (Apache-2.0); ImageNet non-commercial research only; CLIP MIT; all-MiniLM-L6-v2 Apache-2.0; sd-turbo Stability AI non-commercial licence (comparison only). The model card is in the repo.

Limits of the approach: 1000 fixed classes, 128 px, no attributes or composition; mapping quality depends on the LLM; the CLIP score is itself biased by CLIP's web training data.